

# QECTOR Workbench - Complete API Reference

**\*\*Workbench 0.5.2 - Backend qector\_decoder\_v3 0.7.0 (min 0.7.0) - 56 MCP tools - 16 decoders - 10 code families\*\***

**\*\*Decoder package: qector\_decoder\_v3 0.7.0 (bundled wheel, activated offline on first launch)\*\***

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This manual is generated from the live application source so every tool name, decoder kind, code family, and function signature matches the running build exactly.

## Code families

| Family             | Parameter       | Type | Graphlike | Decoders                                                | Notes                                                  |
|--------------------|-----------------|------|-----------|---------------------------------------------------------|--------------------------------------------------------|
| repetition         | distance        | int  | yes       | all (16)                                                | 1D chain parity-check code.                            |
| ring               | distance        | int  | yes       | all (16)                                                | Periodic 1D chain.                                     |
| rotated_surface    | distance        | int  | yes       | all (16)                                                | Standard rotated surface code.                         |
| unrotated_surface  | distance        | int  | yes       | 15 (lookup_table refused >20 checks)                    | Square lattice surface code.                           |
| toric              | distance        | int  | yes       | 15 (lookup_table refused >20 checks)                    | Toric code with periodic boundaries.                   |
| heavy_hex          | distance        | int  | yes       | all (16)                                                | IBM heavy-hex lattice.                                 |
| hypergraph_product | distance        | int  | yes       | all (16)                                                | CSS from repetition seed; graphlike.                   |
| bicycle            | circulant size  | int  | no        | all (16)                                                | qLDPC bicycle code; graphlike enough for all decoders. |
| bivariate_bicycle  | preset index    | int  | no        | 13 (excludes union_find, fast_union_find, lookup_table) | IBM bivariate bicycle code.                            |
| color_code         | triangular size | int  | no        | colour_code, bp_osd, blossom, hybrid, auto_router       | Triangular & 2D surface codes.                         |

## Decoder kinds

| Kind                           | Description                                                                       | Options                          | Compatibility        |
|--------------------------------|-----------------------------------------------------------------------------------|----------------------------------|----------------------|
| union_find                     | Fast approximate matching via union-find.                                         | bp_method, osd_order ignored     | graphlike only       |
| fast_union_find                | Faster union-find variant; approximate, higher LER.                               | -                                | graphlike only       |
| blossom                        | Weight-optimal exact MWPM; matches PyMatching LER.                                | -                                | all                  |
| sparse_blossom                 | Region-growing near-optimal matching; not exact.                                  | -                                | graphlike only       |
| sparse_blossom_radix_neighbors | RadixHeap k-NN candidate edge discovery variant of region-growing matching.       | -                                | graphlike only       |
| bp_osd                         | Belief propagation + ordered statistics for LDPC/qLDPC.                           | bp_method, osd_order, error_rate | all                  |
| auto                           | Self-selecting AutoDecoder.                                                       | -                                | graphlike only       |
| hybrid                         | Combines multiple strategies; chooses per problem.                                | -                                | graphlike only       |
| lookup_table                   | Exhaustive syndrome-to-correction table; refused above 20 checks.                 | -                                | small codes only     |
| predecoded                     | Fast pre-decoding pass before matching.                                           | -                                | graphlike only       |
| auto_router                    | Policy decoder: matching for graphlike, bp_osd for qLDPC. Universally compatible. | -                                | all                  |
| hybrid_cascade                 | Union-Find pre-filter + Blossom/BP-OSD escalation; exposes cascade stats.         | escalation, err                  |                      |
| gnn_belief_matching            | GNN-guided weighted matching with faithfulness fallback.                          | gnn_hidden_size, gnn_n_laye      |                      |
| belief_matching                | BP posteriors reweight exact Blossom matching; faithfulness fallback.             | bp_method, osd_ord               |                      |
| two_stage                      | Two-stage decode pipeline (fast stage + exact escalation).                        | escalation                       | graphlike only       |
| ambiguity_cluster              | Cluster decoding for high-noise/non-graphlike degenerate syndromes.               | -                                | non-graphlike        |
| colour_code                    | Native colour-code decoder over undecomposed detector error models.               | -                                | color_code family on |

## Decoder options

| Key               | Type   | Applies to              | Description                                                       |
|-------------------|--------|-------------------------|-------------------------------------------------------------------|
| bp_method         | string | bp_osd, belief_matching | "exact" or "min_sum".                                             |
| osd_order         | int    | bp_osd, belief_matching | 0, 1, or 2. Higher is slower/more accurate.                       |
| error_rate        | float  | all                     | Physical error probability used to weight edges or set BP priors. |
| escalation        | string | hybrid_cascade          | "blossom" or "bp_osd".                                            |
| max_accept_weight | int    | hybrid_cascade          | Maximum syndrome weight accepted by the pre-filter.               |
| gnn_hidden_size   | int    | gnn_belief_matching     | Hidden dimension of the GNN.                                      |
| gnn_n_layers      | int    | gnn_belief_matching     | Number of GNN message-passing layers.                             |

Unknown keys are ignored with a warning; missing keys use backend defaults.

## Measured data

All figures below were measured on this machine (seeded, n=50, p=0.05, rotated\_surface d=5). They are workload- and hardware-dependent.

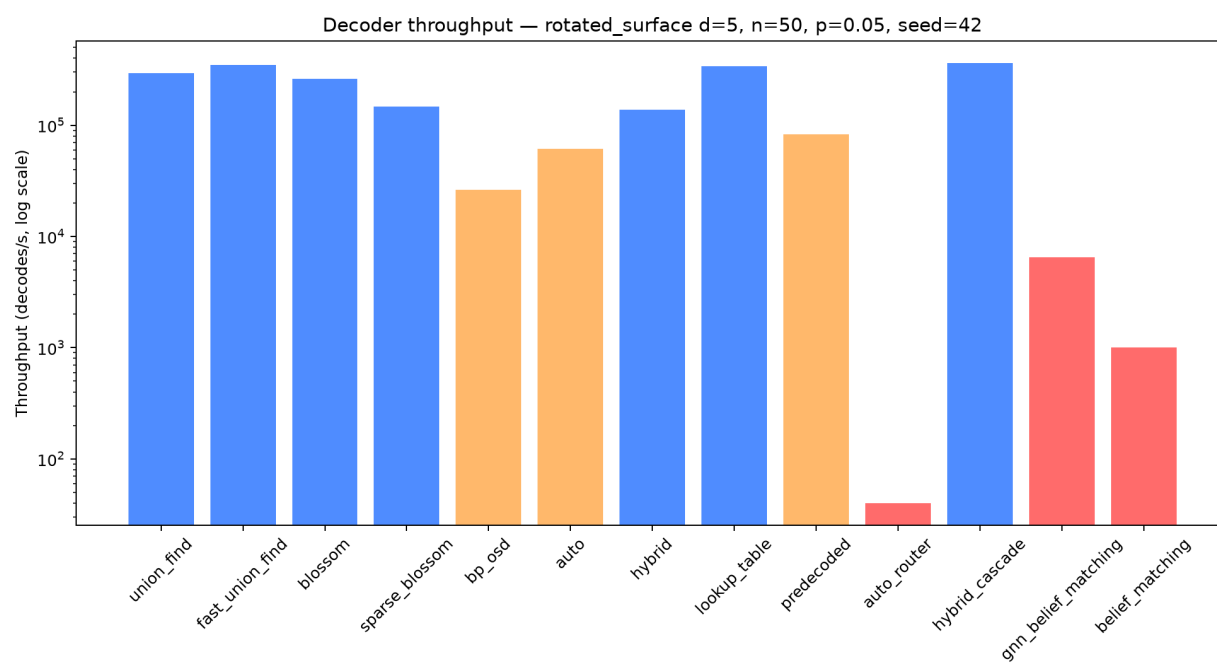
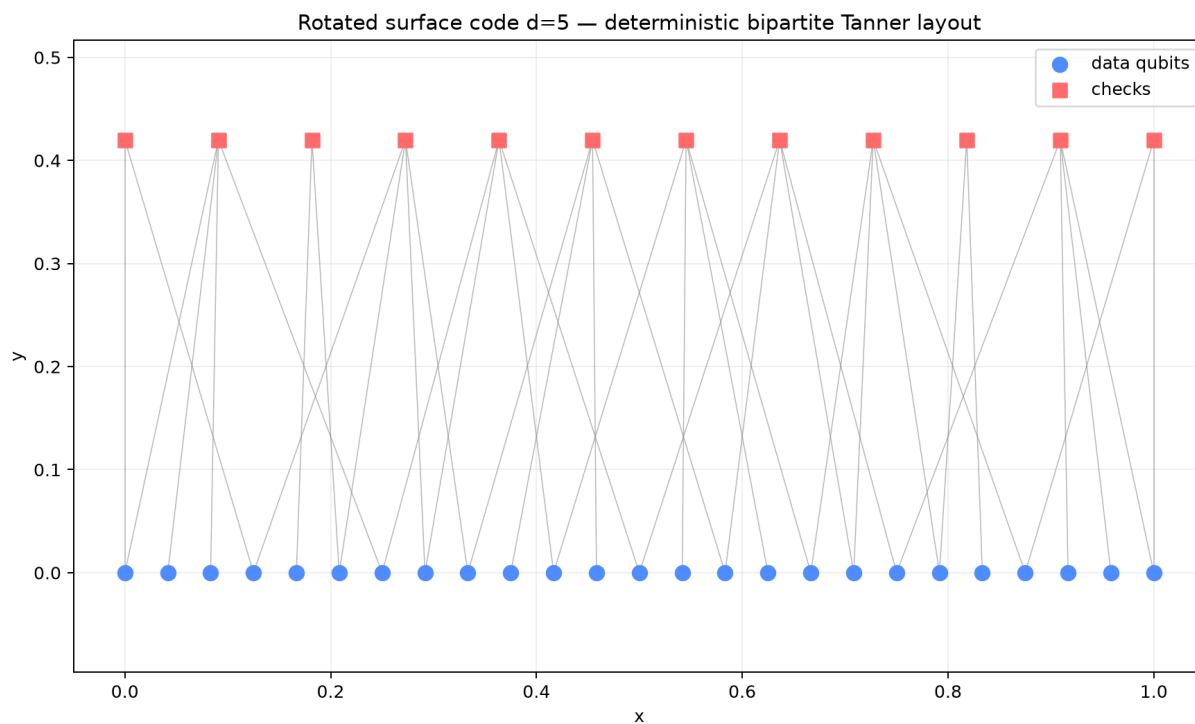
## Code family properties

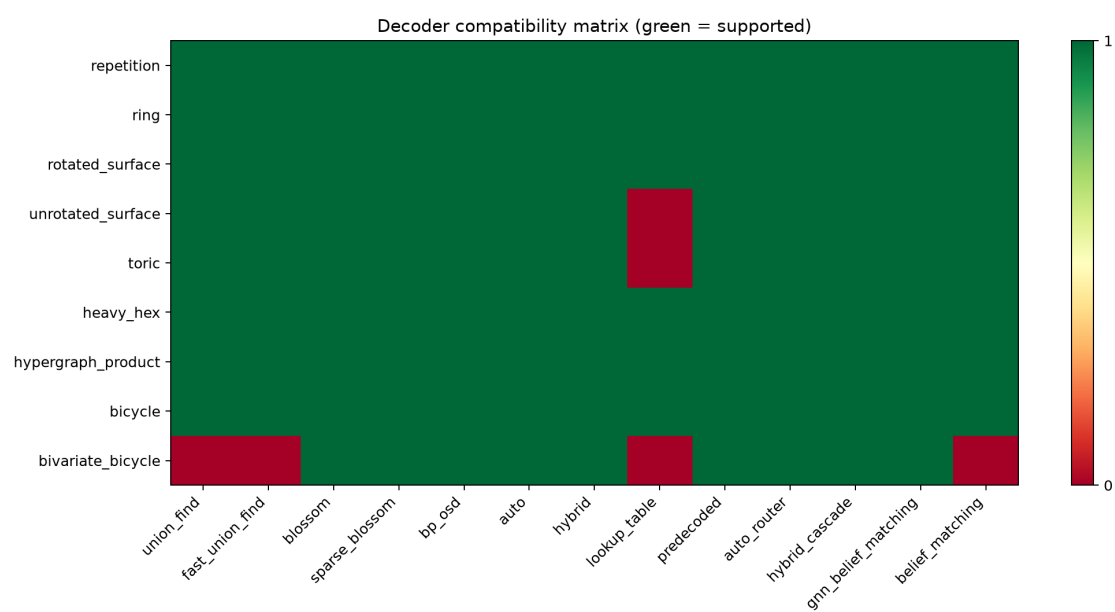
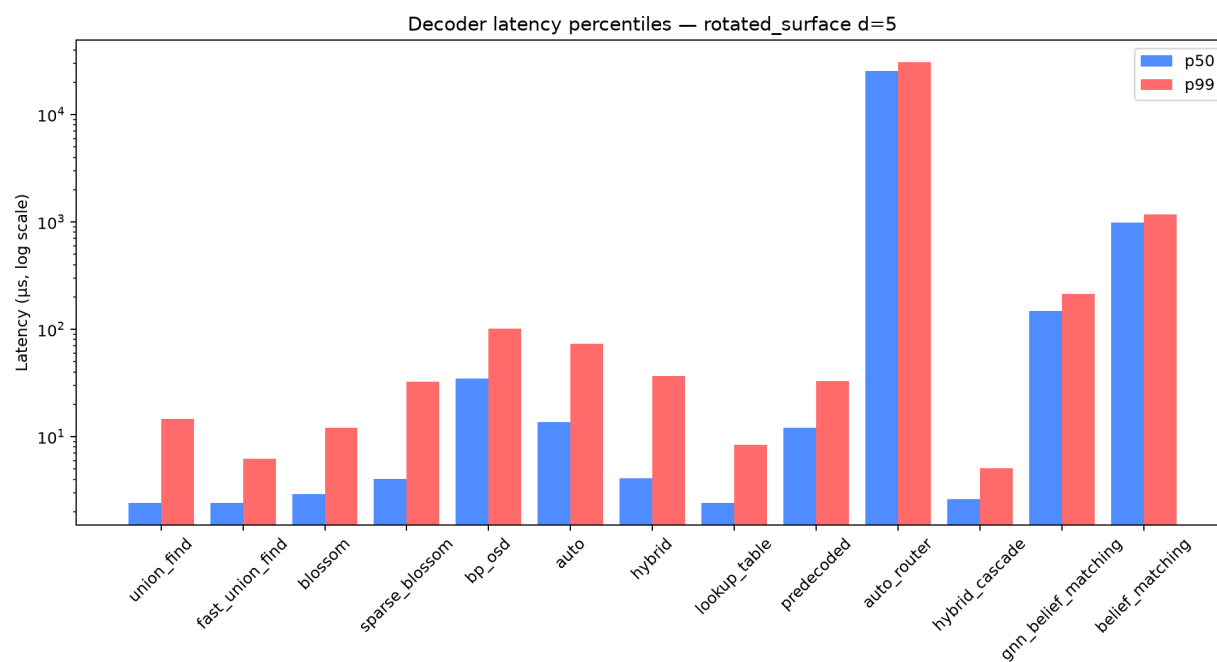
| Family             | Distance | n_qubits | n_checks | max_degree | compatible decoders |
|--------------------|----------|----------|----------|------------|---------------------|
| repetition         | 5        | 5        | 4        | 2          | 13                  |
| ring               | 5        | 5        | 5        | 2          | 13                  |
| rotated_surface    | 5        | 25       | 12       | 2          | 13                  |
| unrotated_surface  | 5        | 40       | 25       | 2          | 12                  |
| toric              | 5        | 50       | 25       | 2          | 12                  |
| heavy_hex          | 5        | 25       | 20       | 2          | 13                  |
| hypergraph_product | 5        | 41       | 20       | 2          | 13                  |
| bicycle            | 5        | 10       | 5        | 2          | 13                  |
| bivariate_bicycle  | 3        | 72       | 36       | 3          | 9                   |

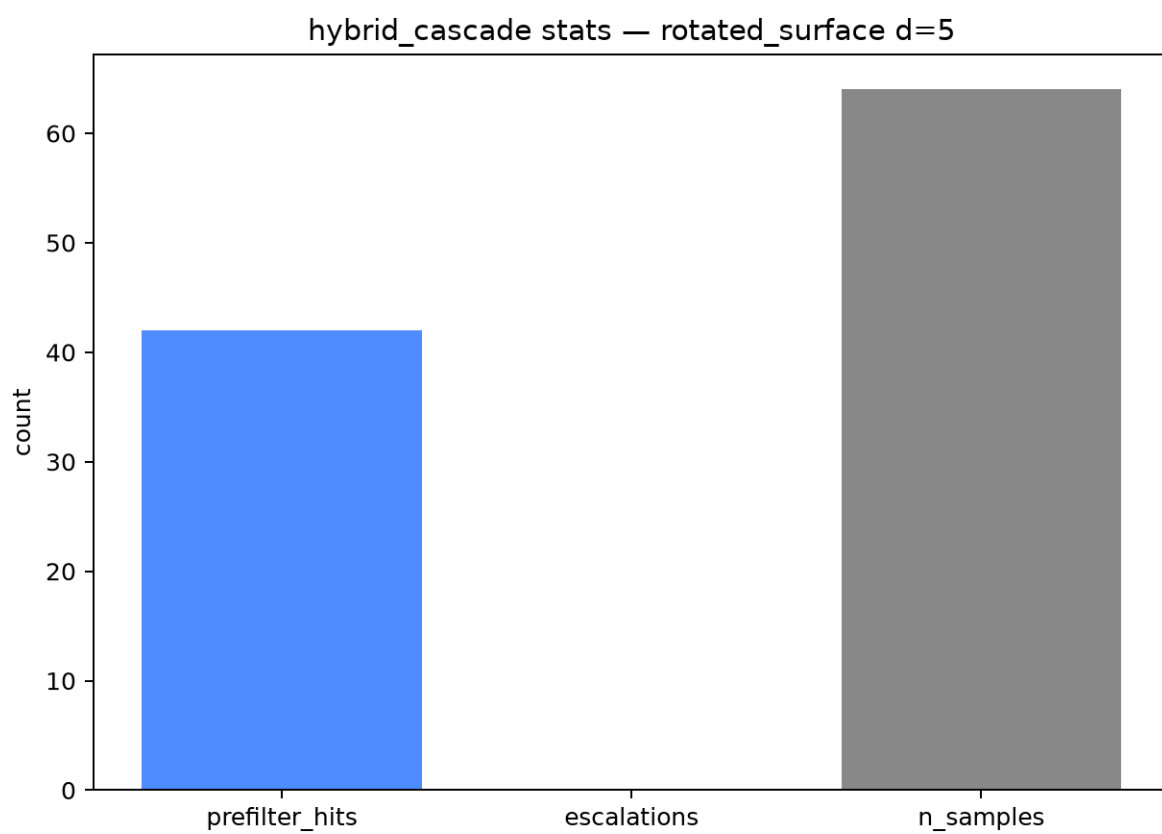
## Decoder benchmark results

| Decoder             | Throughput (decodes/s) | p50 latency (μs) | p99 latency (μs) | logical error rate |
|---------------------|------------------------|------------------|------------------|--------------------|
| union_find          | 295508                 | 2.40             | 14.54            | 0.1                |
| fast_union_find     | 349895                 | 2.40             | 6.25             | 0.1                |
| blossom             | 261917                 | 2.90             | 12.02            | 0.08               |
| sparse_blossom      | 146757                 | 4.05             | 32.63            | 0.08               |
| bp_osd              | 26162                  | 34.75            | 101.55           | 0.1                |
| auto                | 61125                  | 13.60            | 73.22            | 0.1                |
| hybrid              | 138812                 | 4.10             | 36.70            | 0.08               |
| lookup_table        | 337610                 | 2.40             | 8.43             | 0.1                |
| predecoded          | 82850                  | 12.00            | 32.98            | 0.08               |
| auto_router         | 40                     | 25457.50         | 30758.21         | 0.08               |
| hybrid_cascade      | 362845                 | 2.60             | 5.05             | 0.1                |
| gnn_belief_matching | 6520                   | 147.15           | 213.77           | 0.08               |
| belief_matching     | 1001                   | 988.05           | 1173.28          | 0.02               |

## Figures







## backend.py API

**``backend.Any(*args, **kwargs)``**

Special type indicating an unconstrained type.

**``backend.Optional(*args, **kwargs)``**

Optional[X] is equivalent to Union[X, None].

**``backend.QectorError(...)``**

Raised for invalid operations in the QECTOR backend.

**``backend.build_code(family_key: 'str', param: 'int')``**

Build a code from a family and parameter (distance).

**``backend.clear_decoder_cache() -> 'bool'``**

Clear the native decoder cache in qector\_decoder\_v3.

**``backend.code_summary(code) -> 'dict[str, Any]``**

Return a summary dict for a code object.

**``backend.compat_report() -> 'dict[str, Any]``**

Ecosystem-integration availability report.

**``backend.compatible_decoder_kinds(code) -> 'list[str]``**

Return the decoder kinds that can actually construct on code.

**``backend.compute_spring_layout(n_qubits: 'int', n_checks: 'int', check_matrix, iterations: 'int' = 100) -> 'tuple[list, list]``**

Backwards-compatible alias for :func:compute\_tanner\_layout.

**``backend.compute_tanner_layout(n_qubits: 'int', n_checks: 'int', check_matrix) -> 'tuple[list, list]``**

Deterministic bipartite Tanner-graph layout.



**`backend.deque(...)`**

deque([iterable[, maxlen]]) --> deque object

**`backend.flush\_usage(customer\_id: 'Optional[str]' = None, api\_key: 'Optional[str]' = None) -> 'dict[str, Any]`**

Flush accumulated usage metrics to Stripe metered billing API.

**`backend.get\_code\_family\_info(family\_key: 'str') -> 'dict[str, str]`**

Return metadata about a code family.

**`backend.get\_compatible\_decoders(code) -> 'list[dict[str, str]]`**

Return decoder info for the decoders that can construct on this code.

**`backend.get\_decoder\_info(kind: 'str') -> 'dict[str, str]`**

Return human-readable info about a decoder kind.

**`backend.get\_tanner\_graph\_layout(code, family: 'str', distance: 'int') -> 'tuple[list, list]`**

Return qubit and check coordinates for a clean bipartite Tanner graph.

**`backend.list\_available\_codes() -> 'dict[str, Any]`**

Code families wired into the workbench plus the backend's native codes.list\_codes() catalogue (v0.6.6). Pure introspection.

**`backend.logical\_failure(logicals: 'np.ndarray', error, correction) -> 'bool`**

Public: True iff residual (error+correction)%2 flips a logical.

**`backend.logicals\_matrix(code) -> 'Optional[np.ndarray]`**

Public accessor for the code's logical-operator matrix (or None).

**`backend.make\_decoder(code, decoder\_kind: 'str', decoder\_options: 'Optional[dict]' = None) -> 'Any`**

Public: construct a decoder of decoder\_kind for code.

**`backend.native\_recommend(family\_key: 'Optional[str]' = None, distance: 'Optional[int]' = None, n\_qubits: 'Optional[int]' = None, priority: 'str' = 'balanced', batch\_size: 'int' = 1) ->**

**'dict[str, Any]'**

Backend-native decoder recommendation (v0.6.6 recommend).

**`backend.run\_batch\_decode(code, backend: 'str' = 'cpu', n\_samples: 'int' = 100, error\_rate: 'float' = 0.05, seed: 'int' = 1) -> 'dict[str, Any]'**

Run a batch decode on the given code.

**`backend.run\_benchmark(code, n\_samples: 'int' = 1000, seed: 'int' = 42, decoder\_kind: 'str' = 'union\_find', error\_rate: 'float' = 0.05) -> 'dict[str, Any]'**

Run a decode benchmark on the given code.

**`backend.run\_diagnostic\_decode(code, error\_rate: 'float' = 0.05, decoder\_kind: 'str' = 'blossom', seed: 'int' = 42) -> 'dict[str, Any]'**

Rich single decode via v0.6.6 decode\_with\_diagnostics.

**`backend.run\_doctor\_checks() -> 'dict[str, Any]'**

Run system health diagnostic checks via qd.doctor.

**`backend.run\_hybrid\_cascade\_stats(code, n\_samples: 'int' = 64, error\_rate: 'float' = 0.05, seed: 'int' = 1, escalation: 'Optional[str]' = None) -> 'dict[str, Any]'**

Batch-decode with HybridCascadeDecoder and expose its cascade statistics.

**`backend.run\_native\_streaming(code, n\_rounds: 'int' = 8, error\_rate: 'float' = 0.03, seed: 'int' = 1, window\_size: 'int' = 4) -> 'dict[str, Any]'**

Native sliding-window streaming decode (v0.6.6 sliding\_window\_decode).

**`backend.run\_neural\_predecoder\_training(code, n\_samples: 'int' = 200, n\_epochs: 'int' = 5, error\_rate: 'float' = 0.05, seed: 'int' = 1) -> 'dict[str, Any]'**

Train the NeuralPredecoder on sampled (syndrome, error) pairs (lab tool).

**`backend.run\_parallel\_batch\_decode(code, n\_samples: 'int' = 64, error\_rate: 'float' = 0.05, seed: 'int' = 1, decoder\_type: 'str' = 'union\_find', n\_workers: 'Optional[int]' = None) -> 'dict[str, Any]'**

Multi-process parallel batch decode via v0.6.6 DecoderPool.

```
`backend.run_single_decode(code, error_rate: 'float', decoder_kind: 'str', seed: 'int',  
decoder_options: 'Optional[dict]' = None) -> 'dict[str, Any]`
```

\*No docstring.\*

```
`backend.run_streaming_session(code, window_size: 'int' = 5, n_rounds: 'int' = 10,  
error_rate: 'float' = 0.03, seed: 'int' = 1, decoder_kind: 'str' = 'union_find') -> 'dict[str, Any]`
```

Run a sliding-window streaming decode session.

```
`backend.sample_error_and_syndrome(code, error_rate: 'float', seed: 'int')`
```

Public: sample one seeded error and its syndrome for code.

```
`backend.set_license_key_file(path: 'str') -> 'bool`
```

Set license key file path for offline hard-gated verification.

```
`backend.sparse_blossom_radix_neighbors(code_or_checks, defects: 'list[int]', k: 'int' =  
8) -> 'list[tuple[int, int, int, int]]`
```

Return k-nearest candidate edges (sorted by distance) for defects via SparseBlossom RadixHeap.

```
`backend.validate_parameter(family_key: 'str', param: 'int') -> 'tuple[bool, str]`
```

Validate a code family parameter (distance).

```
`backend.verify_correction(code, syndrome, correction) -> 'bool`
```

Public: True iff correction reproduces the observed syndrome.

```
`backend.verify_license_token(token: 'str') -> 'dict[str, Any]`
```

Verify an Ed25519 signed license token string.

## **autodebug.py API**

```
`autodebug.Optional(*args, **kwds)`
```

Optional[X] is equivalent to Union[X, None].

```
`autodebug.asdict(obj, *, dict_factory=)`
```

Return the fields of a dataclass instance as a new dictionary mapping field names to field values.

```
`autodebug.dataclass(cls=None, /, *, init=True, repr=True, eq=True, order=False, unsafe_hash=False, frozen=False, match_args=True, kw_only=False, slots=False, weakref_slot=False)`
```

Add dunder methods based on the fields defined in the class.

```
`autodebug.field(*, default=, default_factory=, init=True, repr=True, hash=None, compare=True, metadata=None, kw_only=)`
```

Return an object to identify dataclass fields.

```
`autodebug.probe_decoders(family: 'str', distance: 'int', error_rate: 'float' = 0.05, seed: 'int' = 42, verify: 'bool' = True) -> 'dict'`
```

Try every wired decoder against one seeded syndrome and report results.

```
`autodebug.resilient_batch_decode(family: 'str', distance: 'int', backend: 'str' = 'cuda', n_samples: 'int' = 100, error_rate: 'float' = 0.05, seed: 'int' = 1, fallback_chain: 'Optional[list]' = None) -> 'dict'`
```

Batch-decode, falling back cuda → opencl → cpu on unavailability.

```
`autodebug.resilient_single_decode(family: 'str', distance: 'int', error_rate: 'float' = 0.05, decoder: 'str' = 'union_find', seed: 'int' = 42, fallback_chain: 'Optional[list]' = None, verify: 'bool' = True) -> 'ResilientDecodeResult'`
```

Decode one seeded syndrome, falling back through decoders on failure.

```
`autodebug.run_self_diagnostics(probe_family: 'str' = 'repetition', probe_distance: 'int' = 3) -> 'DiagnosticsReport'`
```

Run a full environment/decoder/hardware self-test.

```
`autodebug.self_diagnostics(probe_family: 'str' = 'repetition', probe_distance: 'int' = 3) -> 'DiagnosticsReport'`
```

Run a full environment/decoder/hardware self-test.

## **hardware\_routing.py API**

**`hardware\_routing.Optional(\*args, \*\*kwargs)`**

Optional[X] is equivalent to Union[X, None].

**`hardware\_routing.dataclass(cls=None, /, \*, init=True, repr=True, eq=True, order=False, unsafe\_hash=False, frozen=False, match\_args=True, kw\_only=False, slots=False, weakref\_slot=False)`**

Add dunder methods based on the fields defined in the class.

**`hardware\_routing.detectHardware() -> 'HardwareProfile'`**

Detect real GPU/CUDA/OpenCL availability via the installed package.

**`hardware\_routing.opencl\_host() -> 'tuple[int, Optional[str]]'`**

Return (device count, first platform name) exposed by the \*host\* OpenCL.

**`hardware\_routing.opencl\_reason(decoder\_opencl: 'bool', host\_devices: 'int', host\_platform: 'Optional[str]') -> 'str'`**

Explain the OpenCL state in terms a user can act on.

**`hardware\_routing.recommend(code\_family: 'Optional[str]', distance: 'Optional[int]', n\_qubits: 'Optional[int]', priority: 'str') -> 'Recommendation'`**

Heuristic decoder recommendation (deterministic, no model call).

## **version\_service.py API**

**`version\_service.Callable(\*args, \*\*kwargs)`**

Deprecated alias to collections.abc.Callable.

**`version\_service.Optional(\*args, \*\*kwargs)`**

Optional[X] is equivalent to Union[X, None].

**`version\_service.effective\_app\_version(prefer\_latest: 'bool' = False) -> 'Optional[str]'`**

The version the app presents as \*its own\* — the workbench baseline.

**`version\_service.format\_version\_banner(report: 'Optional[dict]' = None) -> 'str'`**

One-line human banner: workbench version + installed backend version.

```
`version_service.get_app_version_info(refresh: 'bool' = False) -> 'dict[str, Any]`
```

Local baseline for the workbench application (offline — no PyPI).

```
`version_service.get_backend_version_info(refresh: 'bool' = False) -> 'dict[str, Any]`
```

Installed backend version, resolved locally — bundled wheel only, no PyPI.

```
`version_service.get_version_report(refresh: 'bool' = False) -> 'dict[str, Any]`
```

Combined app + backend version report (offline — bundled wheel only).

```
`version_service.installed_backend_version() -> 'Optional[str]`
```

The version of the compiled decoder backend actually imported, if any.

```
`version_service.is_newer(latest: 'Optional[str]', current: 'Optional[str]') -> 'bool`
```

True iff latest is a strictly newer version than current.

```
`version_service.local_app_version() -> 'str`
```

The workbench's baked-in baseline version (offline-safe).

```
`version_service.parse_version(v: 'Optional[str]') -> 'tuple`
```

Parse a version string to a tuple of ints for ordering.

```
`version_service.resolve_versions_async(callback: 'Optional[Callable[[dict], None]]' =  
None, refresh: 'bool' = False) -> 'threading.Thread`
```

Resolve both versions on a daemon thread, then invoke callback(report).

## **decoder\_provisioner.py API**

```
`decoder_provisioner.Callable(*args, **kwargs)`
```

Deprecated alias to collections.abc.Callable.

```
`decoder_provisioner.Optional(*args, **kwds)`
```

Optional[X] is equivalent to Union[X, None].

**`decoder\_provisioner.abi\_tag() -> 'str'`**

A short tag unique to this interpreter's binary ABI.

**`decoder\_provisioner.activate\_site() -> 'Optional[Path]'`**

Place the active managed site first on sys.path (idempotently).

**`decoder\_provisioner.active\_site() -> 'Optional[Path]'`**

Read the active managed-site pointer; malformed pointers are ignored.

**`decoder\_provisioner.bootstrap(on\_log: 'Optional[Callable[[str], None]]' = None) -> 'dict'`**

Blocking pre-import gate used by main.py before backend imports.

**`decoder\_provisioner.ensure(prefer\_latest: 'bool' = True, timeout: 'int' = 300, on\_log: 'Optional[Callable[[str], None]]' = None, target\_version: 'Optional[str]' = None) -> 'dict'`**

Ensure an importable decoder, preferring the bundled wheel (offline).

**`decoder\_provisioner.ensure\_async(prefer\_latest: 'bool' = True, callback: 'Optional[Callable[[dict], None]]' = None, on\_log: 'Optional[Callable[[str], None]]' = None, target\_version: 'Optional[str]' = None) -> 'threading.Thread'`**

Run :func:ensure on a daemon thread; upgrade takes effect next launch.

**`decoder\_provisioner.ensure\_dependencies(on\_log: 'Optional[Callable[[str], None]]' = None) -> 'dict'`**

Check and automatically install any missing core dependencies via pip.

**`decoder\_provisioner.find\_local\_wheels() -> 'list[Path]'`**

Find local or bundled .whl files for qector\_decoder\_v3.

**`decoder\_provisioner.import\_ok() -> 'bool'`**

True iff the decoder *actually imports* in this interpreter — i.e. its compiled extension loads. A metadata-only presence check is not enough: a wheel built for another Python ABI leaves valid dist-info but an unloadable .pyd/.so. This is the authoritative "is a usable decoder present?" test used by the boot gate.

**`decoder\_provisioner.is\_frozen() -> 'bool'`**

\*No docstring.\*

**`decoder\_provisioner.managed\_root() -> 'Path'`**

Return the app-owned, user-writable, ABI-scoped decoder storage dir.

**`decoder\_provisioner.purge\_outdated\_managed\_sites(minimum\_ver: 'Optional[str]' = None) -> 'list[str]`**

Delete any on-disk managed decoder site versions older than minimum\_ver.

**`decoder\_provisioner.resolve\_pip\_argv() -> 'tuple[Optional[list[str]], str]`**

Find a pip interpreter that can install extension modules for this app.

**`decoder\_provisioner.scan\_version() -> 'Optional[str]`**

Return the active managed version, otherwise a system-installed version.

**`decoder\_provisioner.self\_check() -> 'dict'`**

\*No docstring.\*

## **doc\_generator.py API**

**`doc\_generator.Optional(\*args, \*\*kwds)`**

Optional[X] is equivalent to Union[X, None].

**`doc\_generator.latex\_escape(value: 'Any') -> 'str'`**

Escape LaTeX special characters (\ & % \$ # \_ { } ~ ^) in value.



## MCP tool reference

56 tools via stdio JSON-RPC 2.0.

### ``ambiguity_cluster_decode``

Decode using AmbiguityClusterDecoder for high noise or non-graphlike codes

**\*\*Parameters\*\***

- family (string, default 'rotated\_surface') - One of: repetition, ring, rotated\_surface, unrotated\_surface, toric, heavy\_hex, bicycle, bivariate\_bicycle, hypergraph\_product, color\_code
- distance (integer, default 5) -
- error\_rate (number, default 0.05) -
- ambig\_threshold (number, default 0.5) -
- max\_cluster\_size (integer, default 12) -
- syndrome ([array', 'null'], required) -
- seed (integer, default 42) -

### ``analyze_code_family``

Analyze a code family with an example code instance

**\*\*Parameters\*\***

- family\_name (string, default 'rotated\_surface') - One of: repetition, ring, rotated\_surface, unrotated\_surface, toric, heavy\_hex, bicycle, bivariate\_bicycle, hypergraph\_product, color\_code
- distance (integer, default 5) -

### ``batch_decode``

Batch-decode sampled syndromes on cpu/cuda/opencl via backend.run\_batch\_decode

**\*\*Parameters\*\***

- family (string, default 'rotated\_surface') - One of: repetition, ring, rotated\_surface, unrotated\_surface, toric, heavy\_hex, bicycle, bivariate\_bicycle, hypergraph\_product, color\_code
- distance (integer, default 5) -
- backend (string, default 'cpu') - One of: cpu, cuda, opencl (no silent fallback)
- n\_samples (integer, default 100) -
- error\_rate (number, default 0.05) -
- seed (integer, default 1) -

### ``batch_decode_gpu``

Batch-decode on an explicit compute backend (cpu/cuda/opencl) with honest availability reporting — unavailable GPU backends return status='unavailable' with a reason, never fake results

**\*\*Parameters\*\***

- family (string, default 'rotated\_surface') - One of: repetition, ring, rotated\_surface, unrotated\_surface, toric, heavy\_hex, bicycle, bivariate\_bicycle, hypergraph\_product, color\_code

- distance (integer, default 3) -
- backend (string, default 'cuda') - One of: cpu, cuda, opengl
- n\_samples (integer, default 32) -
- error\_rate (number, default 0.05) -
- seed (integer, default 1) -

## **`belief\_match\_decode`**

Convenience seeded decode pinned to the belief\_matching kind (BP posteriors + exact MWPM with faithfulness fallback)

**\*\*Parameters\*\***

- family (string, default 'rotated\_surface') - One of: repetition, ring, rotated\_surface, unrotated\_surface, toric, heavy\_hex, bicycle, bivariate\_bicycle, hypergraph\_product, color\_code
- distance (integer, default 5) -
- error\_rate (number, default 0.05) -
- seed (integer, default 42) -

## **`benchmark\_decoder`**

Benchmark a decoder on a code family via backend.run\_benchmark (latency percentiles, throughput, logical error rate)

**\*\*Parameters\*\***

- decoder\_name (string, default 'union\_find') - One of: union\_find, fast\_union\_find, blossom, sparse\_blossom, bp\_osd, auto, hybrid, lookup\_table, predecoded, auto\_router, hybrid\_cascade, gnn\_belief\_matching, belief\_matching, two\_stage, ambiguity\_cluster, colour\_code
- code\_family (string, default 'rotated\_surface') - One of: repetition, ring, rotated\_surface, unrotated\_surface, toric, heavy\_hex, bicycle, bivariate\_bicycle, hypergraph\_product, color\_code
- distance (integer, default 5) -
- error\_rate (number, default 0.05) -
- n\_samples (integer, default 100) -
- seed (integer, default 42) -

## **`check\_updates`**

Report whether the installed decoder backend matches the bundled release baseline (offline — no update service)

**\*\*Parameters\*\***

- refresh (boolean, default False) - Accepted for compatibility; resolution is always local

## **`clear\_decoder\_cache`**

Clear the backend's native decoder cache

\*No parameters.\*

## **`clear\_results`**

Clear all stored benchmark results

**\*\*Parameters\*\***

- confirm (boolean, default False) -

## **`colour\_code\_decode`**

Decode color code using BP-OSD over undecomposed detector error model

**\*\*Parameters\*\***

- distance (integer, default 3) -
- max\_iter (integer, default 30) -
- osd\_order (integer, default 0) -
- syndrome ([array', 'null'], required) -
- seed (integer, default 42) -

## **`compare\_benchmarks`**

Compare stored benchmark results side by side (throughput, p99 latency, logical error rate)

**\*\*Parameters\*\***

- benchmarks (array, required) - result\_id values returned by the run\_benchmark tool

## **`compat\_report`**

Report ecosystem-integration availability (stim/sinter/pymatching/qiskit/ldpc) and research components

\*No parameters.\*

## **`compatible\_decoders`**

Live probe: which decoder kinds construct and produce a syndrome-verified correction on this code

**\*\*Parameters\*\***

- family (string, default 'rotated\_surface') - One of: repetition, ring, rotated\_surface, unrotated\_surface, toric, heavy\_hex, bicycle, bivariate\_bicycle, hypergraph\_product, color\_code
- distance (integer, default 3) -

## **`decode\_single`**

Run one seeded decode and report correction weight, syndrome validity and logical failure

**\*\*Parameters\*\***

- family (string, default 'rotated\_surface') - One of: repetition, ring, rotated\_surface, unrotated\_surface, toric, heavy\_hex, bicycle, bivariate\_bicycle, hypergraph\_product, color\_code
- distance (integer, default 5) -
- decoder\_name (string, default 'union\_find') - One of: union\_find, fast\_union\_find, blossom, sparse\_blossom, bp\_osd, auto, hybrid, lookup\_table, predecoded, auto\_router, hybrid\_cascade, gnn\_belief\_matching, belief\_matching, two\_stage, ambiguity\_cluster, colour\_code
- error\_rate (number, default 0.05) -
- seed (integer, default 42) -

## **`decode\_syndrome`**

Decode an explicit 0/1 syndrome (length `n_checks`) with a chosen decoder; `syndrome_valid` is the GF(2) re-check, `logical_failure` is null (no reference error exists, so it is unknowable)

**\*\*Parameters\*\***

- `family` (string, default 'rotated\_surface') - One of: `repetition`, `ring`, `rotated_surface`, `unrotated_surface`, `toric`, `heavy_hex`, `bicycle`, `bivariate_bicycle`, `hypergraph_product`, `color_code`
- `distance` (integer, default 5) -
- `decoder_name` (string, default 'union\_find') - One of: `union_find`, `fast_union_find`, `blossom`, `sparse_blossom`, `bp_osd`, `auto`, `hybrid`, `lookup_table`, `predecoded`, `auto_router`, `hybrid_cascade`, `gnn_belief_matching`, `belief_matching`, `two_stage`, `ambiguity_cluster`, `colour_code`
- `syndrome` (array, required) - 0/1 syndrome bits, length must equal the code's `n_checks`
- `decoder_options` ([`'object'`, `'null'`], required) - Optional per-decoder construction options: `bp_method` (`exact`|`min_sum`), `osd_order` (`0`|`1`|`2`), `error_rate`, `escalation` (`blossom`|`bp_osd`), `max_accept_weight`, `gnn_hidden_size`, `gnn_n_layers`

## **`decode\_with\_options`**

Seeded decode with validated per-decoder construction options (`bp_osd` `bp_method`/`osd_order`, `hybrid_cascade` `escalation`, GNN architecture); reports `options_applied` honestly

**\*\*Parameters\*\***

- `family` (string, default 'repetition') - One of: `repetition`, `ring`, `rotated_surface`, `unrotated_surface`, `toric`, `heavy_hex`, `bicycle`, `bivariate_bicycle`, `hypergraph_product`, `color_code`
- `distance` (integer, default 3) -
- `decoder_name` (string, default 'bp\_osd') - One of: `union_find`, `fast_union_find`, `blossom`, `sparse_blossom`, `bp_osd`, `auto`, `hybrid`, `lookup_table`, `predecoded`, `auto_router`, `hybrid_cascade`, `gnn_belief_matching`, `belief_matching`, `two_stage`, `ambiguity_cluster`, `colour_code`
- `error_rate` (number, default 0.05) -
- `seed` (integer, default 42) -
- `decoder_options` ([`'object'`, `'null'`], required) - Optional per-decoder construction options: `bp_method` (`exact`|`min_sum`), `osd_order` (`0`|`1`|`2`), `error_rate`, `escalation` (`blossom`|`bp_osd`), `max_accept_weight`, `gnn_hidden_size`, `gnn_n_layers`

## **`delete\_resource`**

Delete a resource by ID

**\*\*Parameters\*\***

- `resource_id` (string, required) -
- `confirm` (boolean, default False) -

## **`diagnostic\_decode`**

Rich single decode via the backend's native `decode_with_diagnostics` (`matched_weight`, `backend_used`, `internal_fallback`, `timing`, `logicals`)

**\*\*Parameters\*\***

- `family` (string, default 'rotated\_surface') - One of: `repetition`, `ring`, `rotated_surface`, `unrotated_surface`, `toric`, `heavy_hex`, `bicycle`, `bivariate_bicycle`, `hypergraph_product`, `color_code`
- `distance` (integer, default 5) -

- decoder\_name (string, default 'blossom') - One of: union\_find, fast\_union\_find, blossom, sparse\_blossom, bp\_osd, auto, hybrid, lookup\_table, predecoded, auto\_router, hybrid\_cascade, gnn\_belief\_matching, belief\_matching, two\_stage, ambiguity\_cluster, colour\_code
- error\_rate (number, default 0.05) -
- seed (integer, default 42) -

## **`doctor\_diagnostics`**

Run system health and environment diagnostic checks via qd.doctor

\*No parameters.\*

## **`export\_benchmark`**

Export a stored benchmark result (by result\_id) to the export directory

**\*\*Parameters\*\***

- benchmark\_id (string, required) - result\_id returned by the run\_benchmark tool
- format (string, default 'json') -

## **`flush\_usage`**

Flush usage metrics to Stripe metered billing API

**\*\*Parameters\*\***

- customer\_id ([ 'string', 'null'], required) -
- api\_key ([ 'string', 'null'], required) -

## **`generate\_documentation`**

Generate code documentation files

**\*\*Parameters\*\***

- family\_key (string, default 'ring') - One of: repetition, ring, rotated\_surface, unrotated\_surface, toric, heavy\_hex, bicycle, bivariate\_bicycle, hypergraph\_product, color\_code
- param (integer, default 6) -
- formats (array, default ['json']) - Any of: json, markdown, html, latex, pdf

## **`get\_code\_properties`**

Get properties of a code family

**\*\*Parameters\*\***

- family\_name (string, default 'ring') - One of: repetition, ring, rotated\_surface, unrotated\_surface, toric, heavy\_hex, bicycle, bivariate\_bicycle, hypergraph\_product, color\_code
- distance (integer, default 5) -

## **`get\_config`**

Get current server configuration

\*No parameters.\*

## **`get\_decoder\_info`**

Get information about a decoder

**\*\*Parameters\*\***

- decoder\_name (string, default 'bp\_osd') - One of: union\_find, fast\_union\_find, blossom, sparse\_blossom, bp\_osd, auto, hybrid, lookup\_table, predecoded, auto\_router, hybrid\_cascade, gnn\_belief\_matching, belief\_matching, two\_stage, ambiguity\_cluster, colour\_code

## **`get\_hardware\_info`**

Get hardware/backend availability

\*No parameters.\*

## **`get\_resource`**

Get a specific resource by ID

**\*\*Parameters\*\***

- resource\_id (string, required) -

## **`get\_resources`**

List all resources

\*No parameters.\*

## **`get\_results`**

Get stored benchmark results (most recent first-in order)

**\*\*Parameters\*\***

- limit (integer, default 10) -

## **`get\_statistics`**

Get server statistics

\*No parameters.\*

## **`get\_system\_info`**

Get system information

\*No parameters.\*

## **`gnn\_belief\_match\_decode`**

Convenience seeded decode pinned to the gnn\_belief\_matching kind with optional GNN architecture overrides (gnn\_hidden\_size, gnn\_n\_layers)

**\*\*Parameters\*\***

- family (string, default 'rotated\_surface') - One of: repetition, ring, rotated\_surface, unrotated\_surface, toric, heavy\_hex, bicycle, bivariate\_bicycle, hypergraph\_product, color\_code

- distance (integer, default 5) -

- error\_rate (number, default 0.05) -
- seed (integer, default 42) -
- gnn\_hidden\_size ([integer, 'null'], required) -
- gnn\_n\_layers ([integer, 'null'], required) -

## **`hybrid\_cascade\_stats`**

Batch-decode through the hybrid\_cascade decoder and expose its live cascade statistics (prefilter\_hits, escalations, hit rate, throughput, syndrome-match rate, logical error rate)

**\*\*Parameters\*\***

- family (string, default 'rotated\_surface') - One of: repetition, ring, rotated\_surface, unrotated\_surface, toric, heavy\_hex, bicycle, bivariate\_bicycle, hypergraph\_product, color\_code
- distance (integer, default 3) -
- n\_samples (integer, default 64) -
- error\_rate (number, default 0.05) -
- seed (integer, default 1) -
- escalation ([string, 'null'], required) - One of: blossom, bposd (default: backend's blossom)

## **`list\_clients`**

List registered clients

\*No parameters.\*

## **`list\_code\_families`**

List available code families

\*No parameters.\*

## **`list\_codes`**

List workbench code families plus the backend's native code catalogue

\*No parameters.\*

## **`list\_decoders`**

List available decoders

\*No parameters.\*

## **`list\_tools`**

List all available MCP tools

\*No parameters.\*

## **`mcp\_status`**

Get MCP server status

\*No parameters.\*

## **`native\_recommend`**

Backend-native decoder recommendation (qector\_decoder\_v3.recommend) with the mapped workbench decoder\_kind

**\*\*Parameters\*\***

- family (string, default 'rotated\_surface') - One of: repetition, ring, rotated\_surface, unrotated\_surface, toric, heavy\_hex, bicycle, bivariate\_bicycle, hypergraph\_product, color\_code
- distance (integer, default 5) -
- n\_qubits ([ 'integer', 'null'], required) -
- priority (string, default 'balanced') - One of: balanced, speed, accuracy
- batch\_size (integer, default 1) -

## **`native\_streaming`**

Native hardware-accelerated sliding-window streaming decode (qector\_decoder\_v3.sliding\_window\_decode) with per-round validity + telemetry

**\*\*Parameters\*\***

- family (string, default 'rotated\_surface') - One of: repetition, ring, rotated\_surface, unrotated\_surface, toric, heavy\_hex, bicycle, bivariate\_bicycle, hypergraph\_product, color\_code
- distance (integer, default 5) -
- n\_rounds (integer, default 8) -
- error\_rate (number, default 0.03) -
- seed (integer, default 1) -
- window\_size (integer, default 4) -

## **`neural\_predecoder\_train`**

Train the NeuralPredecoder research/lab MLP on seeded (syndrome, error) pairs and evaluate on a disjoint held-out stream (exact-match, bit accuracy, syndrome validity, LER)

**\*\*Parameters\*\***

- family (string, default 'repetition') - One of: repetition, ring, rotated\_surface, unrotated\_surface, toric, heavy\_hex, bicycle, bivariate\_bicycle, hypergraph\_product, color\_code
- distance (integer, default 3) -
- n\_samples (integer, default 200) -
- n\_epochs (integer, default 5) -
- error\_rate (number, default 0.05) -
- seed (integer, default 1) -

## **`probe\_decoders`**

Probe which decoders produce a valid (syndrome-verified) correction for a code — a self-test across every wired decoder

**\*\*Parameters\*\***

- family (string, default 'rotated\_surface') - One of: repetition, ring, rotated\_surface, unrotated\_surface, toric, heavy\_hex, bicycle, bivariate\_bicycle, hypergraph\_product, color\_code



- distance (integer, default 5) -
- error\_rate (number, default 0.05) -
- seed (integer, default 42) -

## **`recommend\_decoder`**

Recommend a decoder for a code/priority using detected hardware

**\*\*Parameters\*\***

- family (string, default 'rotated\_surface') - One of: repetition, ring, rotated\_surface, unrotated\_surface, toric, heavy\_hex, bicycle, bivariate\_bicycle, hypergraph\_product, color\_code
- distance (integer, default 5) -
- n\_qubits ([integer, 'null'], required) -
- priority (string, default 'balanced') - One of: balanced, speed, accuracy

## **`register\_client`**

Register a client

**\*\*Parameters\*\***

- client\_id (string, required) -
- access\_level (string, default 'USER') -

## **`reset\_config`**

Reset configuration to defaults

**\*\*Parameters\*\***

- confirm (boolean, default False) -

## **`resilient\_decode`**

Single decode with automatic multi-decoder fallback and a full attempt trace (autodebug.resilient\_single\_decode)

**\*\*Parameters\*\***

- family (string, default 'rotated\_surface') - One of: repetition, ring, rotated\_surface, unrotated\_surface, toric, heavy\_hex, bicycle, bivariate\_bicycle, hypergraph\_product, color\_code
- distance (integer, default 5) -
- decoder\_name (string, default 'union\_find') - One of: union\_find, fast\_union\_find, blossom, sparse\_blossom, bp\_osd, auto, hybrid, lookup\_table, predecoded, auto\_router, hybrid\_cascade, gnn\_belief\_matching, belief\_matching, two\_stage, ambiguity\_cluster, colour\_code
- error\_rate (number, default 0.05) -
- seed (integer, default 42) -

## **`run\_benchmark`**

Run a benchmark and store the result under a generated result\_id

**\*\*Parameters\*\***

- code\_family (string, default 'rotated\_surface') - One of: repetition, ring, rotated\_surface, unrotated\_surface, toric, heavy\_hex, bicycle, bivariate\_bicycle, hypergraph\_product, color\_code
- distance (integer, default 5) -
- decoder\_name (string, default 'union\_find') - One of: union\_find, fast\_union\_find, blossom, sparse\_blossom, bp\_osd, auto, hybrid, lookup\_table, predecoded, auto\_router, hybrid\_cascade, gnn\_belief\_matching, belief\_matching, two\_stage, ambiguity\_cluster, colour\_code
- n\_samples (integer, default 100) -
- seed (integer, default 42) -
- error\_rate (number, default 0.05) -

## **`self\_diagnostics`**

Run a full environment/decoder/hardware self-diagnostics report (autodebug.run\_self\_diagnostics)

\*No parameters.\*

## **`set\_config`**

Merge key/value pairs into the server configuration

**\*\*Parameters\*\***

- config (object, required) - Key/value pairs merged into the current configuration

## **`set\_license\_key\_file`**

Set license key file path for offline verification

**\*\*Parameters\*\***

- path (string, required) -

## **`sparse\_blossom\_radix\_neighbors`**

Discover k-nearest candidate edges via SparseBlossom RadixHeap

**\*\*Parameters\*\***

- family (string, default 'rotated\_surface') - One of: repetition, ring, rotated\_surface, unrotated\_surface, toric, heavy\_hex, bicycle, bivariate\_bicycle, hypergraph\_product, color\_code
- distance (integer, default 5) -
- defects ([array, 'null'], required) -
- k (integer, default 8) -

## **`stream\_decode`**

Run a sliding-window streaming decode session via backend.run\_streaming\_session

**\*\*Parameters\*\***

- family (string, default 'rotated\_surface') - One of: repetition, ring, rotated\_surface, unrotated\_surface, toric, heavy\_hex, bicycle, bivariate\_bicycle, hypergraph\_product, color\_code
- distance (integer, default 5) -
- window\_size (integer, default 5) -

- n\_rounds (integer, default 10) -
- error\_rate (number, default 0.03) -
- seed (integer, default 1) -
- decoder\_name (string, default 'union\_find') - One of: union\_find, fast\_union\_find, blossom, sparse\_blossom, bp\_osd, auto, hybrid, lookup\_table, predecoded, auto\_router, hybrid\_cascade, gnn\_belief\_matching, belief\_matching, two\_stage, ambiguity\_cluster, colour\_code

**`two\_stage\_decode`**

Decode using TwoStageDecoder (decoupled X/Z sector decoders)

### **\*\*Parameters\*\***

- family (string, default 'rotated\_surface') - One of: repetition, ring, rotated\_surface, unrotated\_surface, toric, heavy\_hex, bicycle, bivariate\_bicycle, hypergraph\_product, color\_code
- distance (integer, default 5) -
- x\_decoder (string, default 'blossom') -
- z\_decoder (string, default 'blossom') -
- syndrome ([array, 'null'], required) -
- seed (integer, default 42) -

## verify\_license\_token

## Verify an Ed25519 signed license token string

## **\*\*Parameters\*\***

- token (string, required) -

## `version\_info`

App + decoder-backend version report (workbench baseline + installed backend, resolved locally — no network)

**\*\*Parameters\*\***

- refresh (boolean, default False) - Accepted for compatibility; resolution is always local

## MCP wire protocol

Newline-delimited JSON-RPC 2.0 over stdio. Launch with --mcp.

```

{"jsonrpc": "2.0", "id": 1, "method": "initialize", "params": {"protocolVersion": "2024-11-05", "capabilities": {}, "clientInfo": {"name": "SecurityTool", "version": "1.0.0"}}, "result": {"capabilities": {"experimental": {"tools": [{"name": "decode_single", "arguments": {"family": "rotated", "key": "key", "value": "value"}]}, "toolsList": [{"name": "decode_single", "arguments": {"family": "rotated", "key": "key", "value": "value"}]}}, "protocolVersion": "2024-11-05"}, "error": null}
{"jsonrpc": "2.0", "method": "notifications/initialized"}
{"jsonrpc": "2.0", "id": 2, "method": "tools/list"}
{"jsonrpc": "2.0", "id": 3, "method": "tools/call", "params": {"name": "decode_single", "arguments": {"family": "rotated", "key": "key", "value": "value"}}, "result": {"family": "rotated", "key": "key", "value": "value"}, "error": null}

```

Result envelope: `content[0].text` holds a JSON payload; `isError` flags tool-level failure.

## Common result schemas

```
# Single decode result
{"error":[...], "syndrome":[...], "result":{"correction":[...], "hamming_weight":int, "syndrome_valid":bool, "logic":int}}
```

```
# Benchmark result
{"throughput_decodes_per_s":float,"decode_seconds":float,"n_trials":int,"p":float,"seed":int,"method":str,"bac
# Resilient decode
{"success":bool,"used_decoder":str|None,"fallback_used":bool,"syndrome_valid":bool|None,"logical_failure":bool
# Diagnostics report
{"overall_status":"pass|degraded|fail","timestamp":float,"platform":str,"python":str,"workbench_version":str,"
```

## Environment variables

| Variable                                | Effect                                                                            |
|-----------------------------------------|-----------------------------------------------------------------------------------|
| <code>QECTOR_DATA_DIR`</code>           | Relocate all QECTOR user data.                                                    |
| <code>QECTOR_SILENT`</code>             | Set to <code>`1`</code> to suppress the backend startup notice.                   |
| <code>QECTOR_LICENSE`</code>            | Ed25519 token that overrides academic/commercial for testing.                     |
| <code>QECTOR_DISABLE_OPENCL`</code>     | Set to <code>`1`</code> to skip OpenCL probing. It cannot <i>*enable*</i> OpenCL. |
| <code>QECTOR_ENABLE_OPENCL_AUTO`</code> | Allows OpenCL auto-routing, but only when OpenCL is already available.            |

## Provisioning model

The Workbench bundles the decoder wheel inside the application. On first launch it activates qector-decoder-v3 from the bundled wheel into a managed, ABI-scoped user site — fully offline, on every platform. Any outdated managed decoder left by an older release is purged automatically before activation. No internet connection, Python, or pip is required.

A splash screen is shown within roughly a second of launch and closes once the main window is mapped, so the cold start (extracting the wheel on first run, then loading a compiled extension) is never an invisible wait.

## Boot diagnostics

A windowed build has no stderr, so provisioning is logged to files under the per-user data directory:

| File                              | Contents                                                              |
|-----------------------------------|-----------------------------------------------------------------------|
| <code>logs/boot.log`</code>       | Every bootstrap step, the activated site, and the exact import error. |
| <code>logs/boot_stdio.log`</code> | Anything written to stdout/stderr when the build has no console.      |

## Hardware backends

| Backend              | Availability                                                                            |
|----------------------|-----------------------------------------------------------------------------------------|
| <code>cpu`</code>    | Always available.                                                                       |
| <code>cuda`</code>   | Requires an NVIDIA GPU with a healthy driver.                                           |
| <code>opencl`</code> | Reported unavailable by the published wheel, which is built without the OpenCL feature. |

`opencl_is_available()` returning `False` is a property of the decoder build, not of the host:

a machine can expose OpenCL devices and still get `False`, and no environment variable changes

it. `hardware_routing.detect_hardware()` reports `opencl_host_devices` and

openc1\_host\_platform (probed from the host ICD) plus an openc1\_reason string so the two situations can be told apart. Enabling the backend requires rebuilding qector-decoder-v3 with its openc1 Cargo feature.

## Example workflows

```
import backend as be, autodebug
code = be.build_code("rotated_surface", 5)
out = be.run_single_decode(code, error_rate=0.05, decoder_kind="blossom", seed=42)
assert out["result"]["syndrome_valid"]

out2 = be.run_single_decode(code, error_rate=0.05, decoder_kind="bp_osd", seed=7, decoder_options={"bp_method"
probe = autodebug.probe_decoders("bivariate_bicycle", 3, seed=99)
resilient = autodebug.resilient_single_decode("bivariate_bicycle", 3, decoder="union_find", seed=7)
stats = be.run_hybrid_cascade_stats(code, n_samples=64, error_rate=0.05, seed=1)
```